

MINUGA KARTHIK KUMAR

+91-9959186924 | karthiky369@gmail.com | linkedin.com/in/karthik-kumar-minuga | github.com/karthik-kumarux

Software engineer specializing in secure systems development, building automated security tooling — vulnerability assessment platforms, deception systems, and web security auditing tools — using Python, FastAPI, and containerized architectures.

Education

Bachelor of Technology – Computer Science & Engineering (AI & ML)

Madanapalle Institute of Technology & Science

2023 – 2027

CGPA: 8.03 / 10.0

Technical Skills

Programming Languages: Python, JavaScript, C

Backend & Web: Node.js, NestJS, Next.js, REST APIs, JWT Authentication, Flask, Prisma ORM

Databases: PostgreSQL, MongoDB, MySQL, SQLite

Artificial Intelligence: LLM Integration, Tree-sitter, YOLO, Computer Vision, CNNs, DistilBERT, Deep Learning, Model Fine-Tuning

Cybersecurity: Vulnerability Assessment, VAPT, OWASP Top 10, Application Security, Web Security, Attack Surface Analysis, Detection Engineering, Network Security, Threat Hunting, Secure SDLC, DevSecOps, Security Hardening

Security Tools: OWASP ZAP, Nmap, Nikto, Burp Suite, Wireshark, Metasploit, Snort, Suricata, Splunk, ELK Stack, SonarQube, DVWA, pfSense, Autopsy, Kali Linux, Grafana

DevOps & Tooling: Docker, Docker Compose, Nginx, Celery, Redis, Git, VS Code Extension API, Streamlit

Projects

VAPTaaS — Vulnerability Assessment & Penetration Testing Platform

GitHub

Tools | FastAPI | Celery | Redis | PostgreSQL | OWASP ZAP | Nmap | Nikto | React | Claude API

- **Architected** an asynchronous vulnerability assessment platform orchestrating reconnaissance (Nmap), web vulnerability scanning (OWASP ZAP), and misconfiguration checks (Nikto) via Celery task queues, containerized with Docker Compose for repeatable security assessments.
- **Integrated** the Claude API to auto-generate contextual, developer-readable remediation guidance from raw vulnerability output, translating scan findings into actionable security reporting for engineering teams.
- **Implemented** a pytest suite covering scan orchestration and API endpoints, hardening the platform against regressions ahead of deployment.

Distributed Honeypot Platform

GitHub

Tools | Python | FastAPI | Nginx | PostgreSQL | Grafana | Docker

- **Architected** a multi-service deception platform orchestrating independent web, FTP, and API honeypot containers via Docker, fronted by an HTTPS-terminating Nginx reverse proxy to increase realism against reconnaissance and automated attack tools.
- **Engineered** custom web and FTP honeypot services capturing attacker credentials, payloads, source IPs, and user-agent data in real time to support threat detection and attacker behavior analysis.
- **Designed** a PostgreSQL event pipeline normalizing raw activity into structured events, alerts, and attacker profiles, implementing brute-force and scanner detection logic surfaced through Grafana log analysis dashboards.

Security Header Analysis Tool

GitHub

Tools | Python | Streamlit | SQLite | ReportLab | Pandas

- **Engineered** a Streamlit-based web security auditing tool evaluating HTTP response headers against OWASP Top 10 categories, flagging missing/misconfigured controls (CSP, HSTS, X-Frame-Options) to strengthen web application hardening.
- **Implemented** a SQLite-backed scan history engine with rescan-and-compare analysis and concurrent multi-threaded bulk scanning across up to 50 domains, enabling attack surface visibility at scale.
- **Automated** PDF and JSON security reporting using ReportLab, producing color-coded, remediation-oriented assessments for stakeholders.

NexCode — AI-Powered VS Code Extension

GitHub

Tools | TypeScript | Node.js | VS Code Extension API | Tree-sitter | LLM APIs

- **Architected** a graph-based code representation engine using Tree-sitter WASM to compress codebase context, cutting LLM token consumption by up to 60% while preserving semantic accuracy.
- **Engineered** dependency edge mapping to retain cross-file relationships (imports, calls, class hierarchies) inside a compressed context window, enabling reliable AI reasoning over large codebases.
- **Designed** a think → act → observe agent loop with five extensible tools, orchestrating structured, auditable interaction between the LLM and the local development environment.

Crack Detection & Severity Analysis System

GitHub

Tools | Python | YOLO | CNNs | Streamlit

- **Engineered** an AI-based infrastructure inspection pipeline using YOLO object detection and CNNs for automated crack identification and severity classification in civil structures.
- **Built** an end-to-end image processing and automated reporting pipeline, deployed through a Streamlit interface for real-time inspection workflows.

Certifications

- **Cyber Security Professional Program (240 Hrs)** — Ethnotech Academy Pvt. Ltd.; coverage of ethical hacking, network defense, web/API security, SOC operations & SIEM, and cloud security fundamentals; hands-on with Burp Suite, OWASP ZAP, Nmap, Metasploit, Splunk, and Wireshark.
- **Cybersecurity Essentials & Network Basics** — Cisco Networking Academy
- **Fundamentals of Cybersecurity** — Coursera | **Digital 101 Gold Certificate** — NASSCOM

Achievements & Leadership

- Winner, Department-Level Project Expo | HackerRank 3-Star Badge in C Programming (Data Structures & Algorithms)
- Event Coordinator, AIMEX 2K25 — managed end-to-end logistics and participant coordination for a departmental technical fest; Coordinator, IQNITE — designed and ran a competitive quiz platform challenge